

## Chapter-2

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① What is current and current density?

⇒ The amount of electric current, traveling per unit cross-section area is called as current density and expressed in amperes per square meter. The more the current in a conductor, the higher will be the current density.

② Explain Kirchhoff's law?

⇒ (i) The algebraic sum of current meeting at a junction in a circuit is zero. In this law, the current flowing towards the junction are considered as +ve and those flowing away from the junction as -ve currents.

③ What is current?

⇒ Current is a flow of electrical charge carriers, usually electrons or electron-deficient atoms. The common symbol for current is the uppercase letter  $I$ . The standard unit is the ampere, symbolized by  $A$ .

④ What is Wheatstone Bridge principle?

⇒ The Wheatstone bridge works on the principle of null deflection, i.e. the ratio of their resistances are equal

and no current flows through the circuit. Under normal conditions, the bridge is in an unbalanced condition, the bridge is in an unbalanced condition where current flows through the galvanometer.

⑤ What is potential difference?

⇒ The difference in potential between two points that represents the work involved or the energy released in the transfer of a unit quantity of electricity from one point to the other.

⑥ Explain Ohm's law?

⇒ Ohm's law states that the voltage or potential difference between two points is directly proportional to the current or electricity passing through the resistance, and directly proportional to the resistance of the circuit. The

formula for Ohm's law is  $V = IR$ . This relationship between current, voltage and resistance was discovered by German scientist George Simon Ohm. Let us learn more about Ohm's law, Resistance and its applications.

Voltage = Current  $\times$  Resistance.

$V = I \times R$ ; Here,  $V$  = voltage,  $I$  = current and  $R$  = resistance.



The SI Unit of resistance is ohm's and is denoted by  $\Omega$ . This law one of the most basic laws of electricity. It helps to calculate the power, efficiency, current, voltage and resistance of an element of an electrical circuit.

⑦ What is RC circuit?

⇒ A resistor-capacitor circuit (RC circuit) is an electrical circuit consisting of passive components like resistors and capacitors, driven by the current source or the voltage source. The

⑧ Explain charging and discharging for RC circuit?

⇒ Charging for RC circuit:

If a resistor is connected in series with the capacitor forming an RC circuit, the capacitor will charge up gradually through the resistor until the voltage across it reaches that of the supply voltage. The time required for the capacitor to be fully charge is equivalent to about 5 time constants or  $5T$ .

Discharging for RC circuit:

When a voltage source is removed from an RC circuit, the capacitor

or

When an increasing DC voltage is applied to a discharged capacitor, the capacitor draws what is called a "charging current" and "charges up". When this voltage is reduced, the capacitor begins to discharge in the opposite direction.

How we can produce alternating current and e.m.f. from RC circuit?

⇒ An alternating current is produced by switching the potential between two terminals in a fixed time interval.

the frequency. The difference in potential between the positive and negative terminal is expressed in volts.

Internal Battery resistance. If the internal resistance is zero, the terminal voltage equals the emf (electromotive force). In a real battery, there is internal resistance. The terminal voltage,  $\Delta V = \mathcal{E} - Ir$ . The emf is equivalent to the open-circuit voltage.



field within the dielectric.

## Chapter-2

① What is induction? Explain electromagnetic induction?

⇒ ~~Induction is the process for welcoming newly recruited employees and supporting them to adjust to their new roles and working environment.~~

Electromagnetic induction or Induction is a process in which a conductor is put in a particular position and magnetic field is stationary and a conductor is moving. This produces a voltage or EMF (Electromotive force) across the electrical conductor.

Electromagnetic induction phenomena can be explained by a simple example of a coil and a magnet. When a magnet is brought towards a coil, a relative motion is generated between the two due to a magnetic flux.

② What is magnetic field? And magnetic field vector?

⇒ Magnetic Field is the region around a magnetic material or a moving electric charge within which the force of magnetism acts. A pictorial representation of the magnetic field which describes how a magnetic material.

A magnetic field is a vector field that describes the magnetic influence on moving electric charges, electric currents,

and magnetic materials. A moving charge in a magnetic field experiences a force perpendicular to its own velocity and to the magnetic field.

③ How we can calculate magnetic force on a current carrying conductor?

⇒ The magnetic force on a current-carrying wire in a magnetic field is given by  $\vec{F} = I \vec{\ell} \times \vec{B}$ . For part a, since the current and magnetic field are perpendicular in this problem, we can simplify the formula to give us the magnitude and find the direction through the direction through the RHR-1. The angle  $\theta$  is 90 degrees, which means  $\sin \theta = 1$ .

④ How to calculate torque on a current carrying loop?

⇒  $\tau = NIAB \sin \theta$ . This is the torque on a current-carrying loop in a uniform magnetic field. This equation can be shown to be valid for a loop of any shape. The loop carries a current  $I$ , has  $N$  turns, each of area  $A$ , and the perpendicular to the loop makes an angle  $\theta$  with the field  $B$ .



5) Explain Hall - effect and Hall voltage?

⇒ Hall-effect:

The Hall effect is the deflection of electrons (Holes) in an n-type (p-type) semiconductor with current flowing perpendicular to a magnetic field. The deflection of these charged carriers sets up a voltage called the  $V_H$ . The main purpose is of Hall effect to determine if a substance is a semiconductor or an insulator. The nature of the charge carriers can be measured. The formula of Hall-effect is  $V_H = EB$ . Hall-effect sensors do not wear so have a long life and in case of two-part technology, meaning they have a virtually unlimited life. They are highly reliable. Offer pre-programmable electrical angles and outputs. Offer high-speed operation.

Hall-voltage:

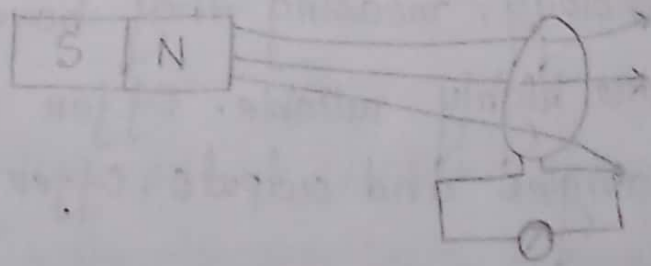
The output voltage, called the hall voltage ( $V_H$ ) of the basic Hall Element is directly proportional to the strength of the magnetic field passing through the semiconductor material (output  $\propto H$ ). The potential difference produced across an electrical conductor when an external magnetic field is applied perpendicular to the current through the conductor.

When calculating the Hall voltage, we need to know the current through the material, the magnetic field, the length, the number of charge carriers and the area. Since all of these are given, the Hall voltage is calculated as:  $V_H = \frac{IB}{nqA}$

In a conductor, the electric field is produced due to the negatively charged free electrons. So, the Hall voltage produced in the conductor is negative.

Q State and explain Faraday's law of electromagnetic induction?

⇒ Law 1: Change of magnetic flux linked to a coil induces an emf across a coil



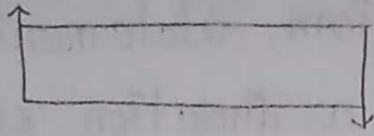
Law 2: The emf induced across the coil is equal to the rate of change of flux in the coil.

$$\epsilon = - \frac{d\Phi_B}{dt}$$

Negative sign shows that emf induced always opposes the change in flux.



## Rotating Magnet



Doubling the speed of rotation will double the emf i.e. Faraday's law is shown to be true.

For leaving certificate physics physics

$$E = \frac{\text{change in } \Phi}{\text{Time}} \quad \text{or} \quad E = \frac{\text{Final flux} - \text{Initial flux}}{\text{time}}$$

⑦ Why Faraday's second law is necessary?

⇒

⑧ What is Lenz Law?

⇒ Lenz's law, in electromagnetism, statement that an induced electric current flows in a direction such that the current opposes the change that induced it. This law was deduced in 1834 by the Russian physicist Heinrich Friedrich Emil Lenz (1804 - 65). Lenz's Law. Related Topics: electromagnetic induction. Lenz's law states that the current induced in the circuit which is due to the change in the magnetic field is directed which opposes the change in the flux and it exerts the mechanical force which opposes the motion.

⑨ Discuss self induction and mutual induction.

⇒ Self inductance is defined as the induction of a voltage in a current-carrying wire when the current in the wire itself is changing. In the case of self-inductance, the magnetic field created by a changing current in the circuit itself induces a voltage in the same circuit. Therefore, the voltage is self-induced.

Mutual Induction:

Mutual induction is defined as the property of the coils that enables it to oppose the changes in the current in another coil. With a change in the current of one coil, the flux changes too thus inducing EMF in the other coil. This phenomenon is known as mutual induction.



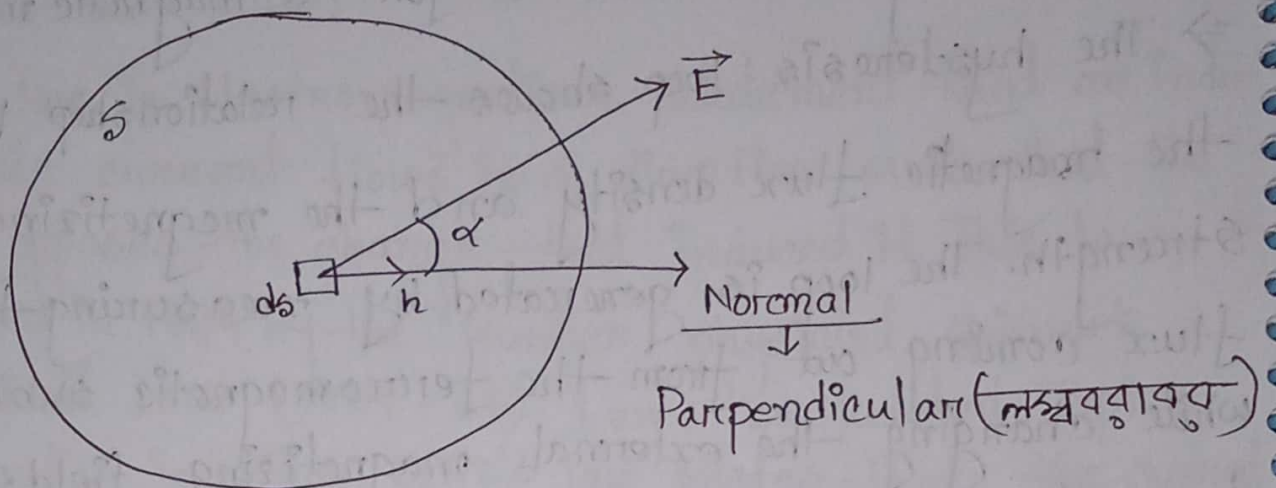
⑩ Explain Hysteresis curve for a magnetic material.

⇒ The hysteresis loop shows the relationship between the magnetic flux density and the magnetizing field strength. The loop is generated by measuring the magnetic flux coming out from the ferromagnetic substance while changing the external magnetizing field. The curvature of the hysteresis is characteristic of the type of material being observed and can vary in size and shape. The basic use of hysteresis curve lies in the selection of suitable materials for different purposes like core of transformer/generator, electromagnets, permanent magnets etc. The choice is made on the basis of properties like retentivity, coercivity, energy loss etc.

### Gauss's Law :

The flux of the electric field  $\vec{E}$  through any closed surface, i.e. the integral  $\oint \vec{E} \cdot d\vec{s}$  over the surface is equal to  $1/\epsilon_0$  times the total charge of the enclosed surface and if the surface does not enclose charge the flux of  $\vec{E}$  i.e.  $\oint \vec{E} \cdot d\vec{s}$  equals zero.

Electric flux over a closed surface :



$$d\phi_E = \vec{E} \cdot \hat{n} dS = \vec{E} \cdot d\vec{S} = E dS \cos \alpha$$

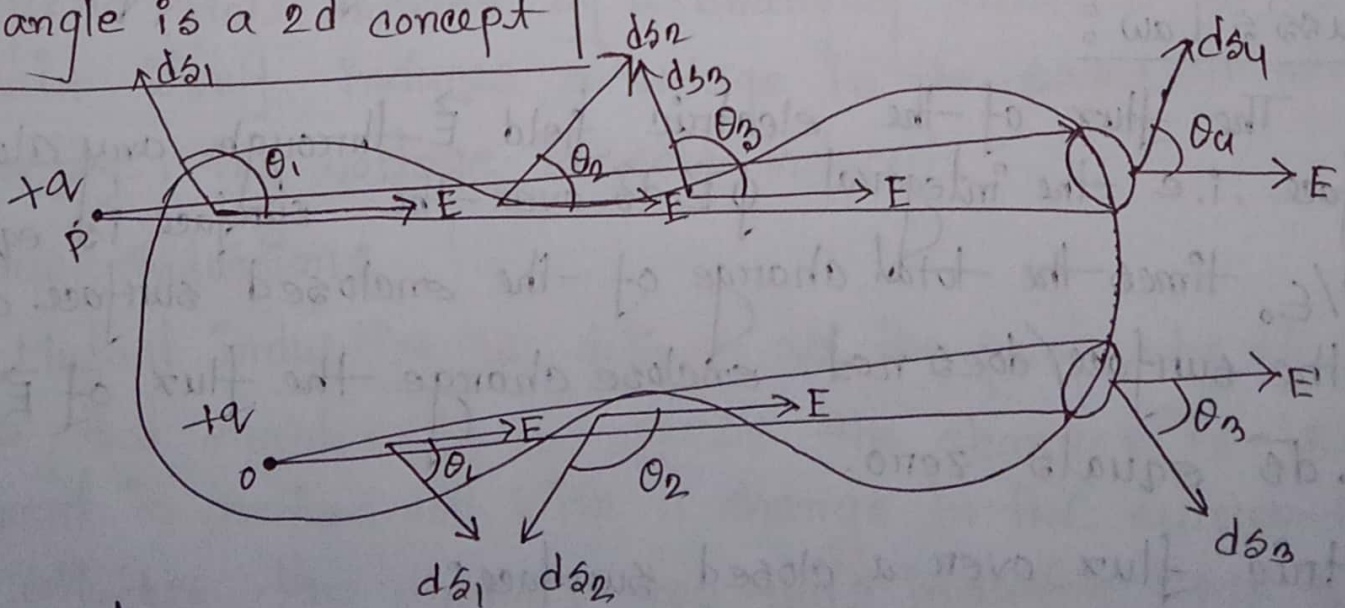
Electric flux over a closed surface can be define as the dot product between electric field intensity and area vector (closed surface area)

$$\boxed{d\phi_E = \oint \vec{E} \cdot d\vec{S}} \rightarrow \text{total electric flux}$$

$$\textcircled{1} \oint \vec{E} \cdot d\vec{S} = 1/\epsilon_0 q \quad [\text{where } q \text{ is the total charge}]$$

$$\text{or, } \oint \vec{E} \cdot d\vec{S} = 0 \quad [\text{if } q \text{ outside the closed surface}]$$

$\#$  angle is a 2d concept



i.e = that is  
at 0 point



Total electric flux,

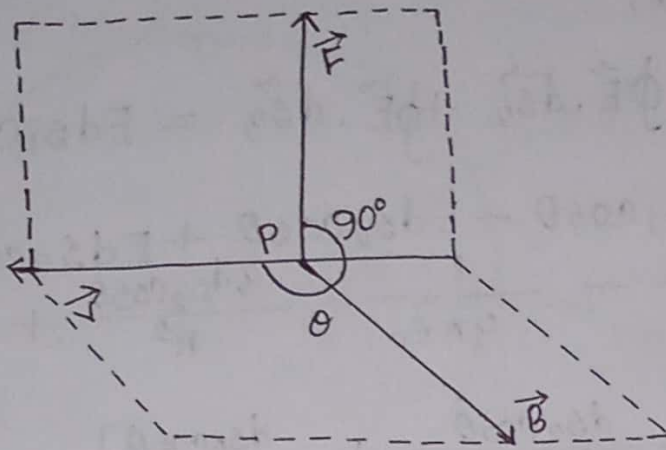
$$\begin{aligned}\Phi_E &= \oint \vec{E} \cdot d\vec{s}_1 + \oint \vec{E} \cdot d\vec{s}_2 + \oint \vec{E} \cdot d\vec{s}_3 = E ds_1 \cos \theta_1 - E ds_2 \cos \theta_2 \\ &+ E ds_3 \cos \theta_3 = E ds_1 \cos \theta - E ds_2 \cos \theta + E ds_3 \cos \theta \\ &= \frac{1}{4\pi\epsilon_0} \frac{q ds_1 \cos \theta}{r^2} - \frac{1}{4\pi\epsilon_0} \frac{q ds_2 \cos \theta}{r^2} + \frac{1}{4\pi\epsilon_0} \frac{q ds_3 \cos \theta}{r^2} \\ &= \frac{q}{4\pi\epsilon_0} \left[ \frac{ds_1 \cos \theta}{r^2} - \frac{ds_2 \cos \theta}{r^2} + \frac{ds_3 \cos \theta}{r^2} \right] \\ &= \frac{q}{4\pi\epsilon_0} [d\omega_1 - d\omega_2 + d\omega_3] \quad [d\omega_1 = d\omega_2 = d\omega_3, \text{ all are solid angle}] \\ &= \frac{q}{4\pi\epsilon_0} [d\omega_1 - d\omega_1 + d\omega_1] = \oint \vec{E} \cdot d\vec{s} = \frac{q}{4\pi\epsilon_0} d\omega \\ &= \oint \vec{E} \cdot d\vec{s} = \frac{q}{4\pi\epsilon_0} \cdot \frac{ds \cos \theta}{r^2} = \frac{q}{\epsilon_0} \cdot \frac{4\pi\epsilon_0}{4\pi\epsilon_0} = q/\epsilon_0\end{aligned}$$

Similarly at p point,

$$\oint \vec{E} \cdot d\vec{s} = 0 \quad [\text{as } q \text{ is outside closed surface}]$$

☐ The magnetic field ( $\vec{B}$ ) vector:

In 1820 Oersted discovered that a current in a wire can also produce magnetic effect. We define, a space around a magnet or a current carrying conductor as the region of magnetic field. The basic magnetic field vector  $\vec{B}$  is called magnetic induction.



If we vary the direction of  $\vec{v}$  through point P, keeping the magnitude  $B$  changes and for  $\vec{v}$  unchanged through  $\vec{F}$  will remain at right angles to  $\vec{v}$ , its magnitude  $F$  changes and for particular orientation the force becomes zero. We define this direction as the direction of  $\vec{B}$ .

In general we can write,  $\vec{F} = q\vec{v} \times \vec{B}$  or  $\vec{F} = qvB \sin\theta$

$$\therefore \frac{|\vec{F}|}{B} = qv \sin\theta \quad \text{--- (1)}$$

$$\vec{B} = \frac{\text{Newton}}{\text{Coulomb m s}^{-1}} = \text{N Coul}^{-1} \text{ m}^{-1} \text{ s} = \frac{\text{N}}{\text{Am}}$$

$$= \text{Weber m}^{-2}$$

$$\text{Weber} = \frac{\text{Nm}}{\text{A}}$$

define 1 weber

### The Lorentz Force:

If a charged potential moves through a region where both an electric & magnetic field are present, then the resultant force is given by,

$$\vec{F} = q_0 \vec{E} + q_0 \vec{v} \times \vec{B} = q_0 [\vec{E} + \vec{v} \times \vec{B}]$$